

SDS 164 Project: ALIENS

HOW DO THEY APPEAR TO US? WILLEMIJN LEINER AND MCKELVEY MCDANIEL

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- So, what kinds of cars *do* they drive?

The aliens are here. We will be examining aliens' different modes of transportation over the years. Using a dataset on UFO sightings from 1949 to 2014, we will inspect their different preferences as to shape, season, duration of appearance, and location. As Sigmond Axel stated, "many of the data records were frightening, and very disturbing leaving a sense of loss and terror even with the short amount of text."¹ In this unprecedented time, we must wonder if extraterrestrial beings can be the cause. By scrutinizing when, where, and how they make themselves known to us, we hope to reveal more about our possible extraterrestrial overlords. We will be looking at a dataset of self-reported UFO sightings around the United States. If they are going to rule us all, we might as

well know what cars they drive. As our world becomes more skeptical of large entities, are the aliens revealing themselves to us at a similar rate?

This data was collected by the National UFO Reporting Center (NUFORC). Individuals were able to self-report sightings of unidentified flying objects on the NUFORC website, which were then time-normalized and geo-coded by Sigmond Axel and made available on GitHub. Each observational unit represents an individual UFO sighting report submitted by members of the public around the world. The dataset contains data collected between 1949 and 2014. We will be examining the date and the time of the sightings, a date variable, ranging from 10/10/1949 20:30 to 9/9/2013 13:30. In addition we will look at the state, since we are only looking at locations in the US. This is a categorical nominal variable, and all states are included. The country variable will also be used, but just to exclude all countries outside of the US. The duration in seconds is a numerical discrete variable, which has a large range, varying from seconds to hours. We will use the duration in seconds for our project, rather than comparing hours and minutes. The shape is a categorical nominal variable, with many different shapes being mentioned. All NA values will be removed, and we will clean and lump rare categories for a more clear analysis.

Examining Shape

Shape	Number of Reports
round	1284
light	939
sharp	638

other	416
fireball	366
multiple	147
changing	104
cigar	94

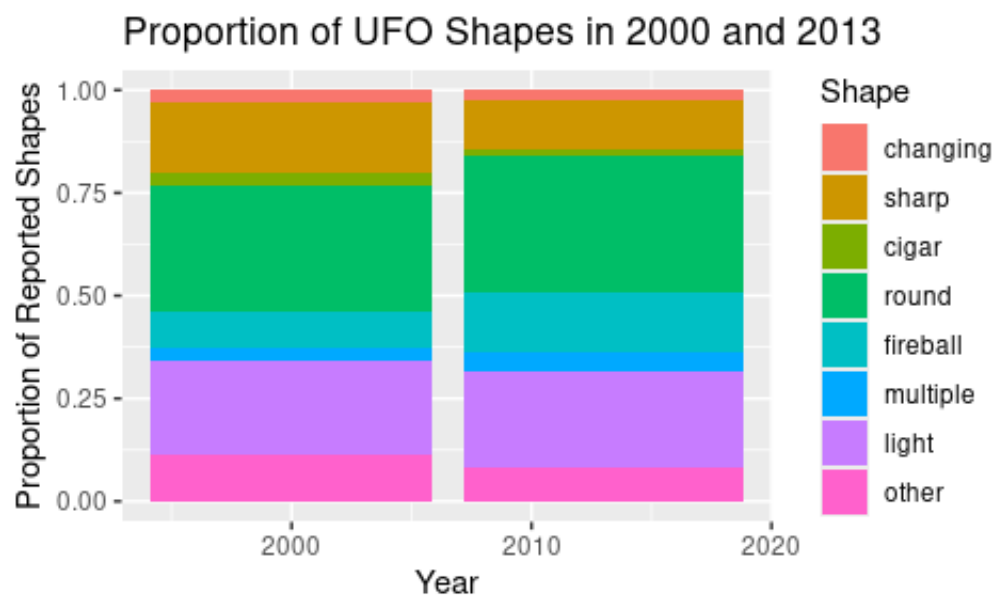


Figure 1. The proportional distribution of the shapes of UFO's reported in the United States remains consistent between 2000 and 2014. Round and light shaped objects make up the largest share of sightings in both of the years. The data includes self reported data on UFO sightings collected by the National UFO Reporting Center, and are filtered to reports from only the years 2000, and 2014, to see if the distribution of UFO shapes had changed between the two years. The bars represent the proportion of total sightings in each year, and are stacked by shape categories. The figure is descriptive and does not include formal statistical testing.

This figure shows that the relative distribution of reported UFO shapes is quite similar in 2000 and 2013. In both of these years, round and light shapes objects make up a large proportion of the UFO sightings, which suggests consistency in how self-reporters describe these UFO sightings over time. There are small changes among the less common shape categories, but there are no big changes in shape are observed. We cannot conclude that the distribution hasn't shifted in the years between 2000 and 2013, but the consistency suggests that changes in the amount of reports over time are not shifting in the types of objects that are being reported.

Deciphering Location

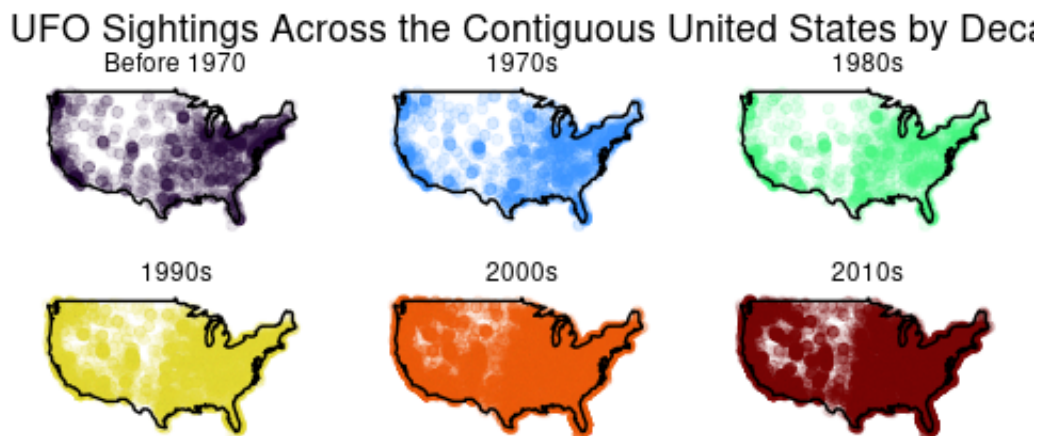


Figure 2. We see that sightings across the country have increased, with consistently more reports in areas with higher populations. Data was collected by the National UFO Reporting Center and includes 61,487 reports.

As we can see, the aliens seem to be revealing themselves to us more and more frequently as the years go by. They are reported more frequently in areas with

more people, not suggesting any favoritism for any specific area.² Because we've only collected information on our extraterrestrial revelations up to 2014, the map representing sightings in the 2010s is partially incomplete. We must consider that UFO reports are filed online and thus have become more accessible with increased access to internet, so it is possible that our alien buddies have been just as present but less reported for decades.

Getting to the Bottom of Dates

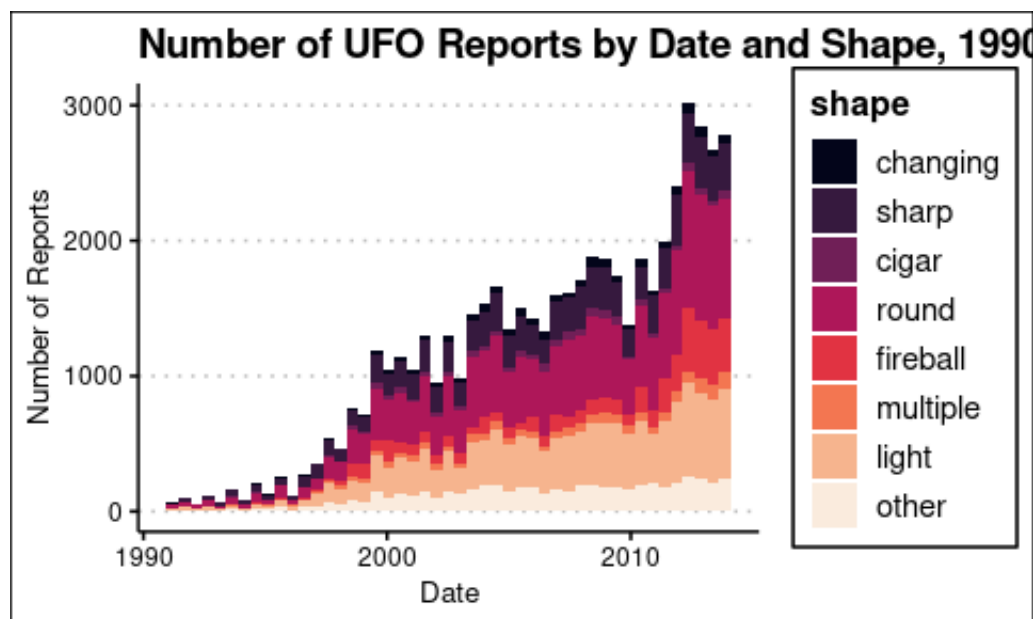


Figure 3. UFO reports have increased from 1990 to 2013, with the largest growth being later in the 1990's, and sightings described as light and round shapes consistently making up a large proportion of reports. The data contains reported UFO sightings in the United States from 1990 to 2013, each report including the date of the sighting and a categorical description of the shape of the object. The y-axis shows the number of reports per time bin, and the x-axis shows the date. The bars are stacked by reported shape category, and no formal statistical tests were conducted.

The stacked histogram shows trend in UFO sightings in the United States over time between 1990 and 2013. The total number of reports have increased majorly over time, particularly in the 1990's, which shows a possible growing interest during this period. Each bar is stacked by UFO shape category, including changing, sharp, cigar, round, fireball, multiple, light, and other. Among most years, light and round shapes have made up a large proportion of reported sightings.

So, what kinds of cars *do* they drive?

It's been made clear that your run-of-the-mill alien is a modern man, appearing late in the evening in something vaguely round and revealing themselves to an equally modern east coast resident. While we can draw some conclusions from our analysis, it is imperative that we acknowledge the limitations inherent with self-reported data. As came up in our preliminary analyses, the nature of an online self-reporting form is that there will be more reports as internet access becomes more standard and more reports in areas with more people to observe extraterrestrial phenomena. We can, however, rest assured that when the aliens undoubtedly come to rule us all, we're better informed about their habits regarding time of day, location, and frequency of appearance.

1. Axel, Sigmond. "Ufo-Reports." GitHub, 16 Sept. 2023, github.com/planetsig/ufo-reports. ↩
2. Alex. "The U.S. Archipelago - Vivid Maps." Vivid Maps, 6 Nov. 2020, vividmaps.com/us-block-level-population-density/#google_vignette. Accessed 28 Jan. 2026. ↩

